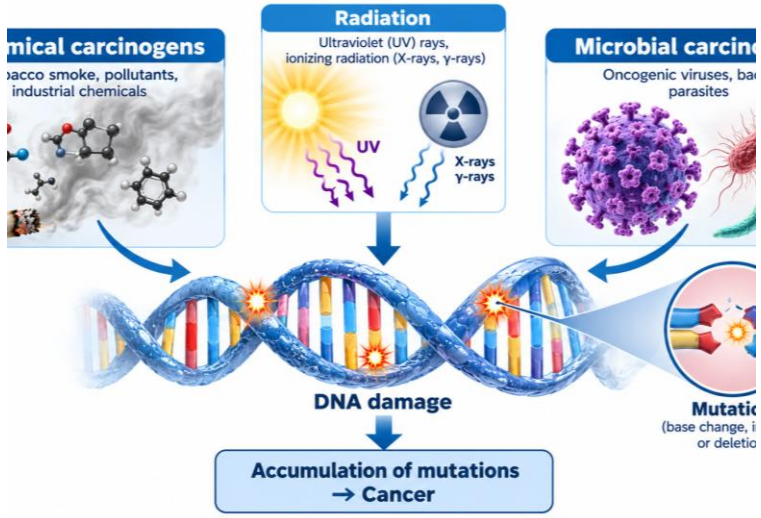
Carcinogenic agents and tumor immunity

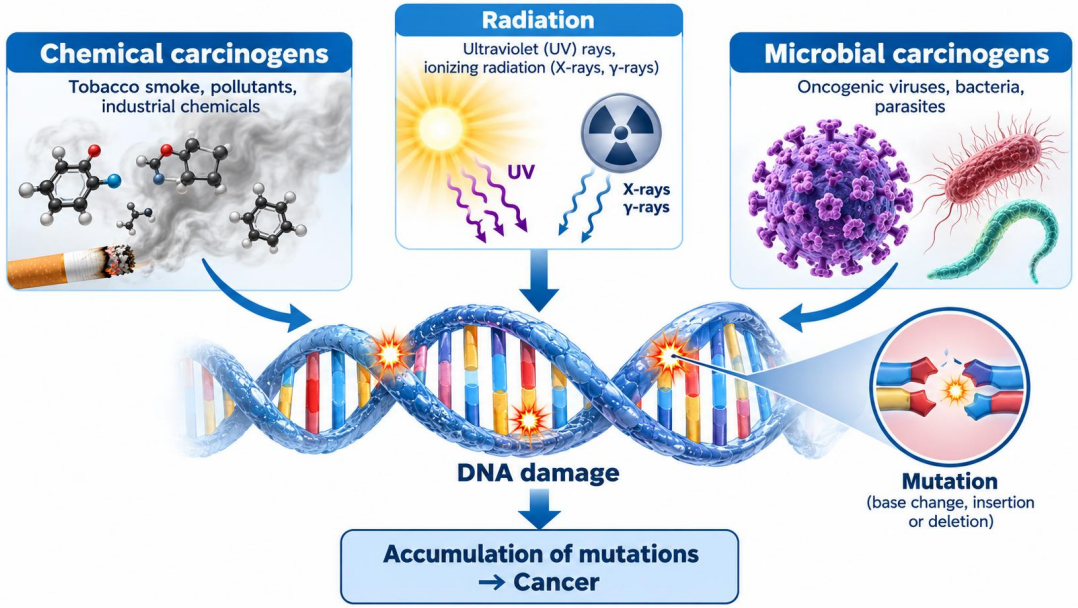
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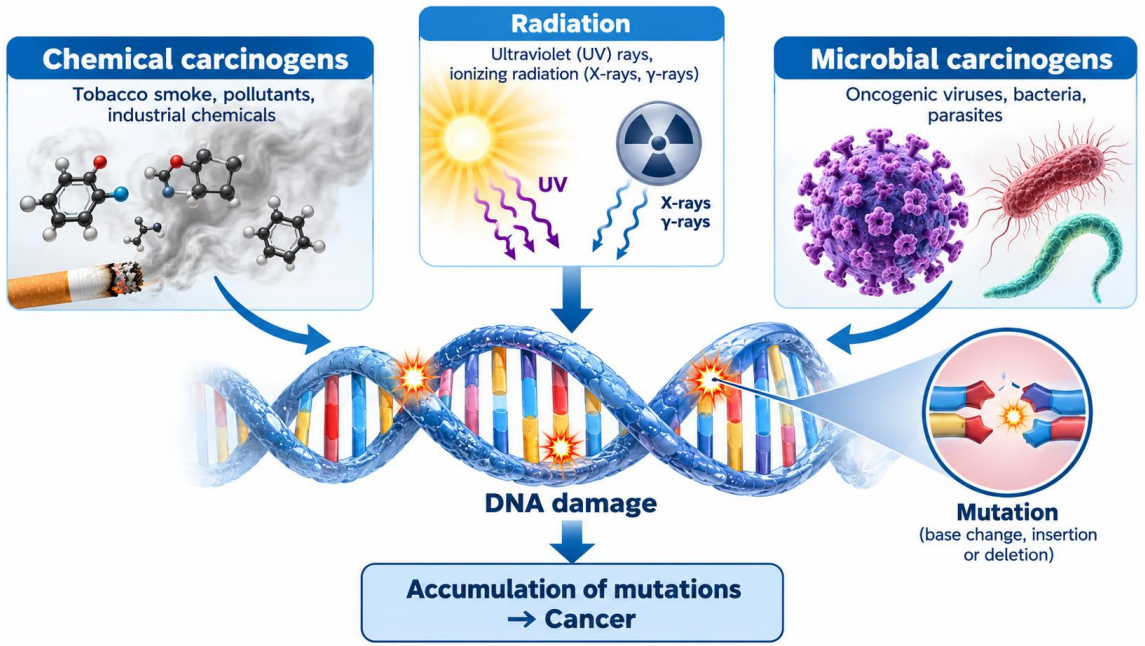


Lecture outlines:

- Carcinogenic agents: chemical, radiation, microbial
- How carcinogens cause DNA damage and mutations
- Tumor immunity: antigens, surveillance, evasion
- Clinical implications: prognosis and immunotherapy



Carcinogenesis



- Carcinogenesis is a multistep process driven by nonlethal genetic damage.
- Agents may directly damage DNA or create chronic inflammation and cell turnover.
- Key targets: oncogenes, tumor suppressor genes, DNA-repair genes, and apoptosis regulators.

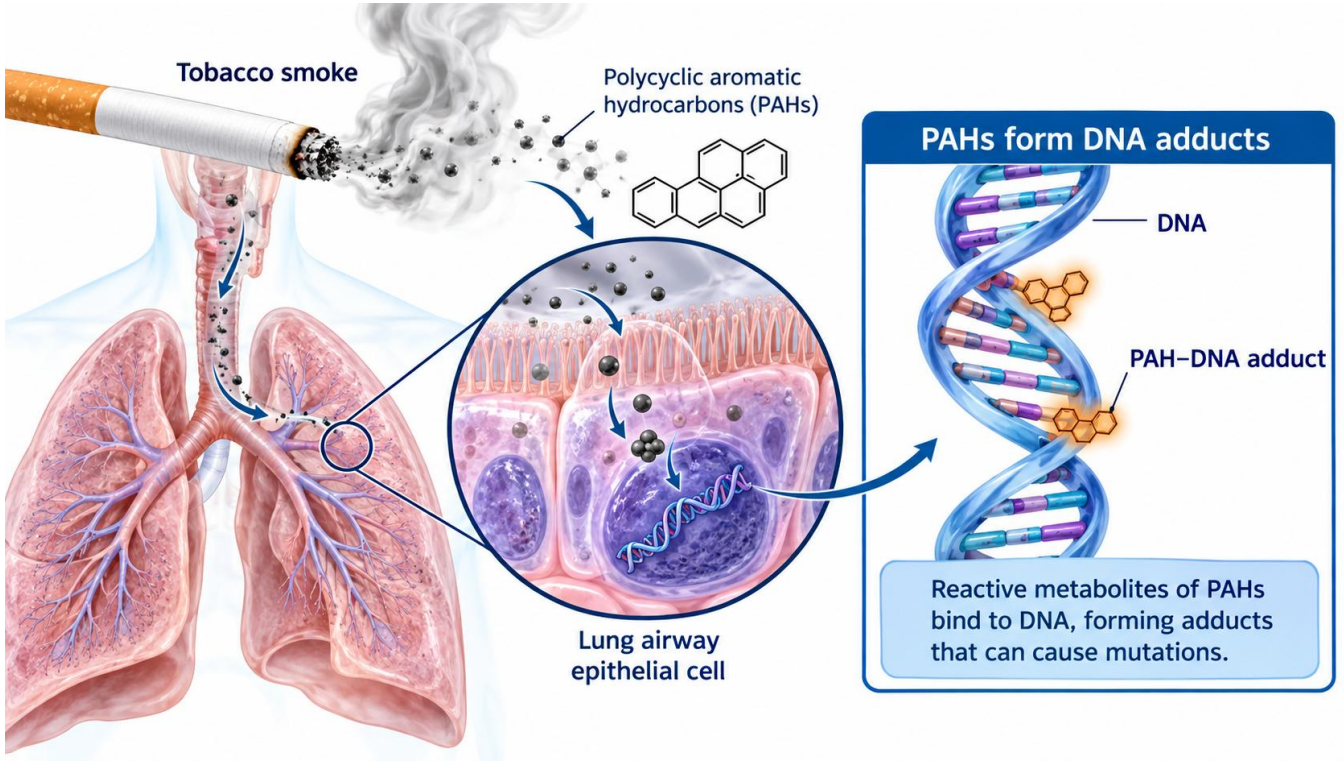
Chemical carcinogens: direct vs indirect

Type	Key point	Examples
Direct-acting	No metabolic conversion needed; generally weak carcinogens	Alkylating chemotherapy drugs
Indirect-acting	Need metabolic activation, often by CYP450 enzymes, to become “ultimate carcinogens”	Polycyclic aromatic hydrocarbons, aromatic amines, nitrosamines, aflatoxin
Mechanism	Form DNA mutations if not repaired	RAS, TP53 and other cancer genes may be affected

Chemical carcinogens

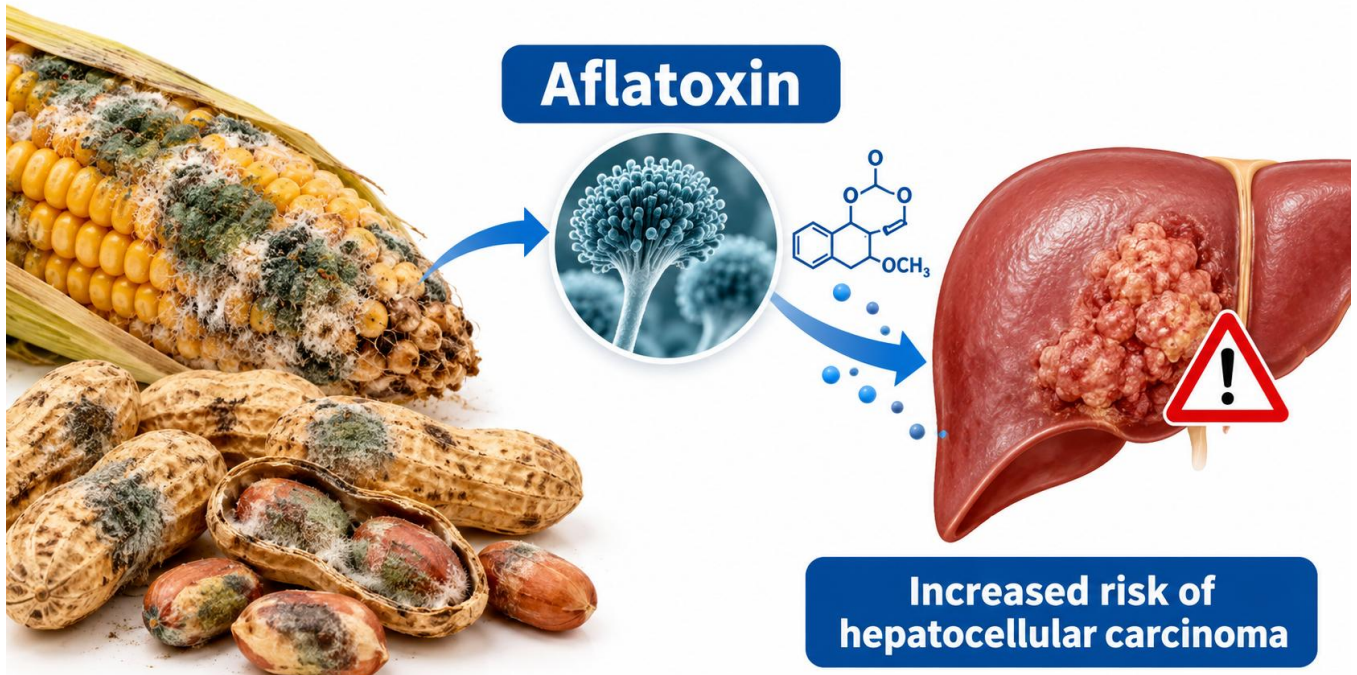
Agent/exposure	Main tumor association
Polycyclic aromatic hydrocarbons (tobacco smoke)	Lung cancer and other smoking-related cancers
Aromatic amines / azo dyes	Bladder cancer
Nitrosamines / amides	Gastric cancer
Asbestos	Mesothelioma; bronchogenic carcinoma
Vinyl chloride	Angiosarcoma of liver
Arsenic	Skin SCC; lung cancer; hepatic angiosarcoma
Aflatoxin B1	Hepatocellular carcinoma

Tobacco smoke: PAHs form DNA adducts



- High-temperature combustion creates PAHs such as benzo[a]pyrene.
- Metabolic activation produces reactive intermediates that bind DNA.
- Unrepaired DNA adducts may result in mutations during replication.

Aflatoxin B1 and liver cancer

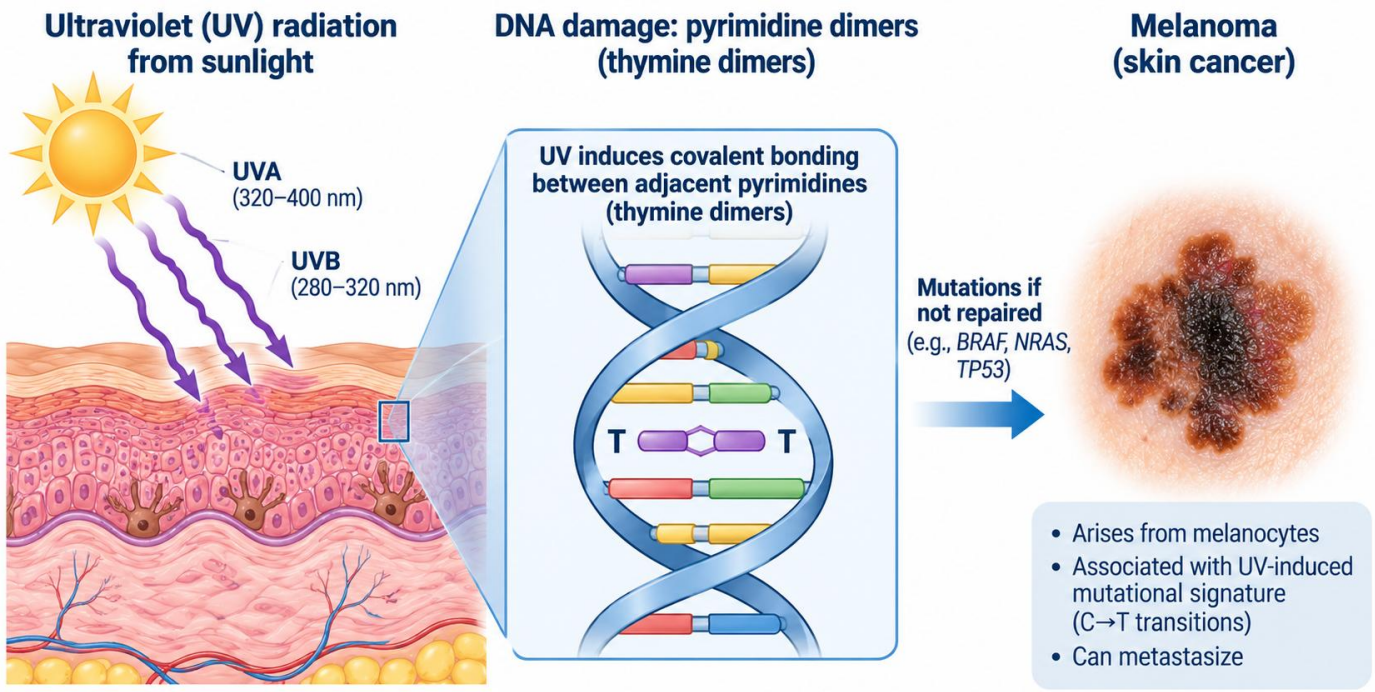


- Produced by Aspergillus species growing on poorly stored grains and nuts.
- Metabolized to a reactive epoxide that causes DNA damage.
- Strongly linked to hepatocellular carcinoma.

Radiation carcinogenesis

Type	DNA injury	Common tumor association
Ultraviolet light	Pyrimidine dimers; faulty repair leads to mutations	Skin SCC, BCC, melanoma
Ionizing radiation	DNA breaks, chromosomal damage, translocations	Leukemia, thyroid cancer, sarcomas
Important note	Tumors may appear years after exposure	Risk depends on dose, age and tissue sensitivity

UV damage → skin cancer

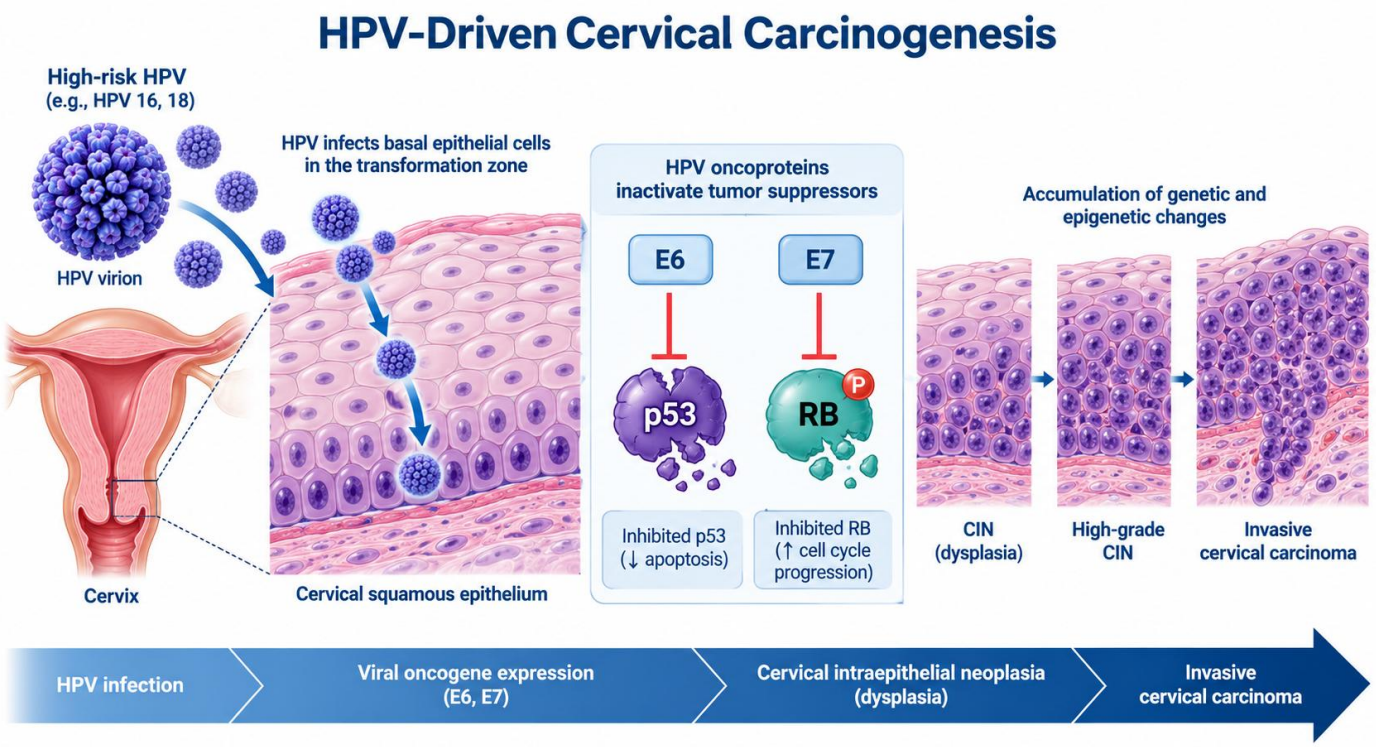


- UVB is the main cause of pyrimidine dimers.
- Repair failure leaves characteristic mutations.
- Outcomes include squamous cell carcinoma, basal cell carcinoma and melanoma.

Microbial carcinogens:

Microbe	Associated cancers
HPV 16/18	Cervical, anogenital, oropharyngeal and laryngeal carcinomas
EBV	Burkitt lymphoma, nasopharyngeal carcinoma, some Hodgkin lymphoma, post-transplant lymphomas
HBV / HCV	Hepatocellular carcinoma
HHV-8 / KSHV	Kaposi sarcoma; primary effusion lymphoma
HTLV-1	Adult T-cell leukemia/lymphoma
H. pylori	Gastric adenocarcinoma; gastric MALT lymphoma
Parasites	Schistosoma haematobium: bladder SCC

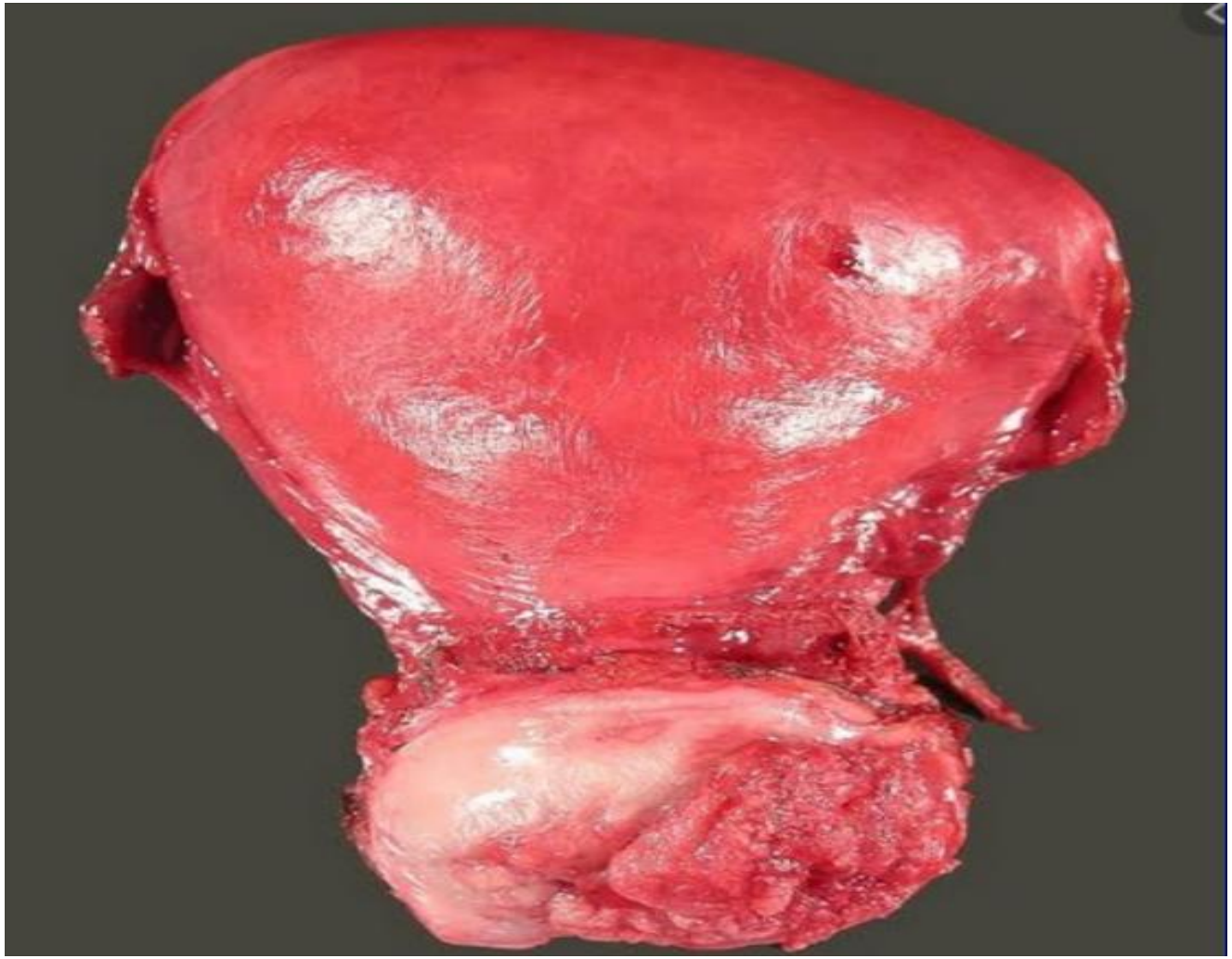
HPV-driven cervical carcinogenesis



- Low-risk HPV 6/11: benign warts.
- High-risk HPV 16/18: major oncogenic types.
- Viral E6 promotes p53 degradation; E7 inactivates RB.
- Result: dysplasia, CIN and invasive carcinoma.



Viral wart caused by low-risk HPV



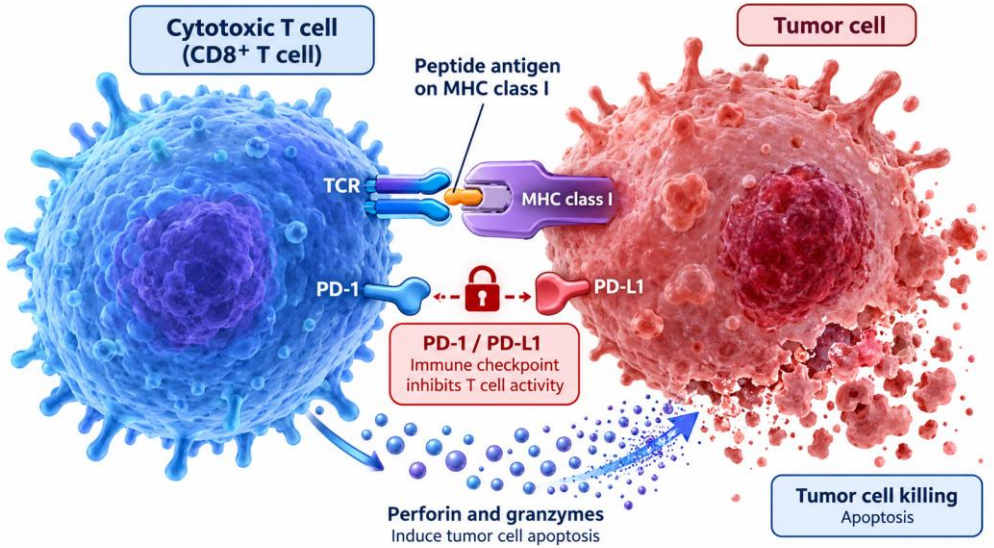
Cervical wart caused by high-risk HPV

Microbial carcinogenesis:

Agent	Mechanism
EBV	B-cell infection and proliferation; cooperates with other events in lymphomas and nasopharyngeal carcinoma
HBV/HCV	Chronic liver injury, regeneration and inflammation create a carcinogenic environment
H. pylori	Chronic gastritis → atrophy/metaplasia/dysplasia; also drives MALT lymphoma
HHV-8	Viral genes promote proliferation and survival in Kaposi sarcoma

Tumor immunity: immune surveillance

- Tumor cells can express antigens recognized as abnormal by the immune system.
- $CD8^+$ cytotoxic T cells are the major anti-tumor effector cells.
- Immune surveillance: the immune system identifies and eliminates emerging malignant cells.



Types of tumor antigens

Antigen type	Example / significance
Neoantigens	New peptides from mutations; often strong targets for T cells
Overexpressed or aberrant normal proteins	e.g., tyrosinase in melanoma
Cancer-testis antigens	Normally germ-cell restricted; expressed in some cancers
Viral antigens	HPV and EBV proteins in virus-associated tumors

Mechanisms of immune evasion

Evasion strategy	How it helps the tumor
Loss of antigen expression	Tumor becomes less visible to T cells
Reduced MHC class I	Impaired antigen presentation
PD-L1 / checkpoint expression	Inhibits T-cell activity
Immunosuppressive microenvironment	TGF-β, regulatory T cells and suppressive cytokines reduce immune attack

Evidence for tumor immunity

- Some tumors contain lymphocytic infiltrates, and stronger immune responses may correlate with better prognosis.
- Occasional spontaneous regression occurs, especially in melanoma and neuroblastoma.
- Cancer risk is increased in immunodeficiency and immunosuppressed patients.
- Tumor-specific antibodies and T-cell responses can be detected in some patients.

Summary:

- Chemical, radiation and microbial carcinogens promote cancer mainly by DNA damage, chronic inflammation or viral oncogene effects.
- High-yield examples: PAHs, aflatoxin, UV light, HPV, EBV, HBV/HCV and H. pylori.
- Tumor immunity depends mainly on $CD8^+$ T cells recognizing tumor antigens.
- Immune evasion, especially checkpoint activation, is clinically targetable.

Thank you

Questions?